THE UNIVERSAL ASKEY-WILSON ALGEBRA MAPS ONTO THE POLYNOMIAL ALGEBRA IN THREE VARIABLES

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Allowed standard prerequisites (editorial).

- The defining relations of the universal Askey-Wilson algebra Delta. Same source, Definition labelled def:uaw; invoked inside the counted proof at lines 2711-2713. Supplies the relations that the three barred generators must satisfy for the homomorphism to exist.
- The generators A, B, C of Delta. Same source, the algebra Delta is generated by A, B, C and the barred images generate the polynomial algebra; used at lines 2714-2731. Gives uniqueness (A, B, C generate the source) and surjectivity (their images generate the target).

Source reference. Original definition 1.2: sources/sigma-2011-069-askey-wilson/pinned-original-source.tex, source lines 2713–2713. Navigation pointer for \ref{def:uaw} used inside the counted span (line 2713); the argument that consumes it is retained in the span.

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Lemma 0.1. *There exists a unique* $\mathbb{F}$*-algebra homomorphism* $\Delta \to \mathbb{F}[\overline{A}, \overline{B}, \overline{C}]$ *that sends*

$$A \mapsto \overline{A}, \qquad B \mapsto \overline{B}, \qquad C \mapsto \overline{C}.$$

This homomorphism is surjective.

Proof. The algebra $\mathbb{F}[\overline{A}, \overline{B}, \overline{C}]$ is commutative, so each of

$$\overline{A} + \frac{q\overline{B}\,\overline{C} - q^{-1}\overline{C}\,\overline{B}}{q^2 - q^{-2}}, \qquad \overline{B} + \frac{q\overline{C}\,\overline{A} - q^{-1}\overline{A}\,\overline{C}}{q^2 - q^{-2}}, \qquad \overline{C} + \frac{q\overline{A}\,\overline{B} - q^{-1}\overline{B}\,\overline{A}}{q^2 - q^{-2}}$$

is central in $\mathbb{F}[\overline{A}, \overline{B}, \overline{C}]$. Consequently $\overline{A}$, $\overline{B}$, $\overline{C}$ satisfy the defining relations for Δ from Definition 1.2. Therefore the homomorphism exists. The homomorphism is unique since A, B, C generate Δ. The homomorphism is surjective since $\overline{A}$, $\overline{B}$, $\overline{C}$ generate $\mathbb{F}[\overline{A}, \overline{B}, \overline{C}]$. □

References

[1] Alperin R.C., Notes: $\mathrm{PSL}_2(\mathbb{Z}) = \mathbb{Z}_2 \star \mathbb{Z}_3$, *Amer. Math. Monthly* **100** (1993), 385–386.